

Whose Family Does “Family-Centered AI” Assume? Reflections from Youth Online Safety Research in Nigeria

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As AI becomes increasingly embedded in family life, emerging approaches to family-centered AI offer opportunities to design beyond the individual. Yet, this shift from individual to family raises questions around *what constitutes a family* and *who are we designing for*? In this position paper, we draw on insights from our research on youth online safety in Nigeria to question assumptions about family structure, technological access, caregiver roles, and privacy that may become embedded in family-centered AI. We argue for a more plural and value-sensitive approach that recognizes families as culturally and materially situated relational systems in which care, authority, privacy, resources, and youth autonomy are continually negotiated.

CCS Concepts: • **Human-centered computing**; • **Social and professional topics** → **Children**;

Additional Key Words and Phrases: Child-Computer Interaction, Marginalized Youth, Participatory Design, Youth Online Safety, Care Work

1 Introduction

Artificial intelligence (AI) is becoming increasingly embedded within everyday family life. Children and caregivers now encounter AI through educational technologies [30], conversational agents [6, 26], recommendation systems [14], and generative AI systems [16]. Research efforts have shown that these technologies do more than support individual users; they can shape how families communicate, make decisions, negotiate rules, distribute responsibilities, and respond to online opportunities and risks [10, 13]. Consequently, there is growing recognition that understanding technologies used by children requires moving beyond the child as an isolated user toward considering the broader family system in which technology use occurs [5, 19, 29]. Family Systems Theory, for example, conceptualizes families as interconnected systems in which the actions and experiences of one member affect others, providing a useful lens for understanding technology as embedded within family relationships rather than as an external influence on individual users [25].

This shift toward family-centered AI creates an important opportunity to reconsider who and what AI systems are designed to support. Yet, we argue that expanding the unit of design from an individual to a family does not necessarily eliminate the normative assumptions embedded within technology. Instead, it risks replacing the idea of a universal user with that of a universal family [18]. What constitutes a family, who performs caregiving, how authority is distributed, who has access to technology, and what privacy means within a household can vary considerably across cultural, socioeconomic, and infrastructural contexts [17]. In this position paper, we draw on insights emerging from our broader research on youth online safety in Nigeria to interrogate assumptions that may become embedded within family-centered AI. Our research does not examine family-centered AI directly. Rather, youth accounts of how they access technology, negotiate online risks, and navigate relationships with caregivers and other people around them provide a useful vantage point for questioning the kinds of families that emerging AI systems might imagine. We argue that family-centered AI should not begin with a predefined model of the family. Instead, it

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should recognize families as culturally and materially situated relational systems in which care, authority, privacy, technological resources, and youth autonomy are continually negotiated.

1.1 The “Assumed Family” in Family-Centered AI

Designing technology for families inevitably requires making decisions about who constitutes the family, what relationships exist among family members, and how technologies should mediate those relationships. Prior research has demonstrated the value of moving beyond individualistic approaches toward understanding families as interconnected systems [5, 19] when understanding youth digital experiences. However, translating this perspective into AI design raises another question: whose family does the system assume? We identify three assumptions that warrant particular attention as family-centered AI develops.

1.1.1 The Family as a Parent-Child Dyad. Technologies intended to support children’s safety frequently position the parent-child relationship as the primary unit of intervention [1, 12]. Parental controls, monitoring applications, content restrictions, and other safety mechanisms commonly assign caregivers the role of protecting or regulating children’s technological experiences [11, 27]. While parents can play important roles in children’s digital lives [28], reducing family-centered design to a parent-child dyad may overlook the broader networks through which caregiving and support occur [3]. Families may include siblings, grandparents, extended relatives, guardians, and other individuals who participate in caregiving and decision-making [2, 5, 17]. Our research has shown that these relationships are even more important for youth from collective cultures such as the Global South [17] or those from marginalized situations [20]. Designing an AI system around a single privileged “parent or youth” account therefore does more than establish an interface hierarchy; it embeds an assumption about who possesses legitimate authority and responsibility within the family. Family-centered AI consequently raises a more fundamental design question than simply how parents and children should share an AI system: Who should be recognized as a caregiver, trusted person, or decision-maker by the system?

1.1.2 Individual Ownership of Devices and Accounts. A second assumption concerns technological access. Contemporary online safety studies are frequently structured around an individual young user possessing a personal device, account, and data profile [9, 24]. AI systems intensify the importance of this assumption because personalization increasingly relies on interaction histories, behavioral data, preferences, and other information associated with an account [4, 31]. Yet access to technology is not uniformly individual. Research across resource-constrained settings has documented practices such as shared-device access, shared connectivity, limited mobile data, and uneven access to reliable digital infrastructure [7, 15, 21]. Such conditions complicate the relationship between an account and the person an AI system believes it represents [22]. For family-centered AI, this produces practical and ethical questions. What does personalization mean when siblings or other family members interact with an AI system through the same device? Whose preferences should the system learn? Thus, infrastructure should not be treated as external to family-centered AI. The material conditions through which families access AI fundamentally shape what kinds of family participation, personalization, and privacy are technologically possible.

1.1.3 Parents as Technologically Equipped Safety Mediators. A third assumption concerns caregivers’ capacity to mediate technology use. Many approaches to children’s online safety depend on caregivers configuring settings, monitoring activity, discussing risks, or responding when problems occur [2, 23]. Such approaches can implicitly presume that caregivers possess sufficient technological literacy, time, connectivity, and familiarity with rapidly evolving platforms to perform this labor effectively. These resources, however, are unevenly distributed. Differences in digital

literacy, education, access, language, time, and familiarity with emerging technologies can influence how caregivers participate in children’s digital lives [23]. Generative AI may make this challenge particularly salient as young people sometimes encounter and appropriate emerging technologies more quickly than the adults responsible for supporting them.

1.2 Reflections from Youth Online Safety Research in Nigeria

Our ongoing research on youth online safety in Nigeria provides a useful standpoint from which to examine these assumptions. As part of Ozioma’s broader dissertation examining how he lived experiences of youth in the global south align with or diverge from dominant understandings of youth online safety, we conducted surveys [15] and qualitative interviews with Nigerian youth about their experiences with digital technologies, online risks, safety practices, and the people and resources they rely upon when navigating online spaces. While this research was not designed specifically to study family-centered AI, participants’ experiences highlight why family and technological context cannot be separated when considering future safety technologies. First, participants’ accounts suggest that online safety can be relational rather than exclusively individual. Young people may draw on parents, siblings, peers, and other trusted individuals when navigating online experiences. At the same time, seeking support does not necessarily imply unrestricted disclosure. Youth may make situated decisions about whom they trust, which experiences warrant assistance, and what information they prefer to manage independently. These practices complicate a model in which an AI system simply detects risk and either flags it or transfers it to a designated moderator. Greater caregiver visibility does not necessarily correspond to greater safety. Instead, AI systems may need to support young people’s ability to determine when assistance is needed and which relationships are appropriate for different circumstances.

Second, participants’ experiences demonstrate how technological access can shape safety practices. Where devices, connectivity, data, or other technological resources are constrained or shared, assumptions about continuous access and individualized technology may not hold. An AI safety system requiring persistent connectivity, individualized accounts, or continuous monitoring could consequently function very differently across households. Finally, our broader findings challenge a purely deficit-oriented conception of youth as passive recipients of technological protection. Youth actively interpret risks, develop strategies, seek assistance, and make decisions about when and how others should become involved in their online experiences. Family-centered AI should therefore consider youth not simply as individuals whom families and AI systems protect, but as stakeholders who participate in defining what safety and appropriate family involvement mean. All these reflections suggest that moving from individual-centered to family-centered AI requires more than expanding the number of stakeholders represented in an interface. It requires understanding the relationships through which safety, care, authority, and autonomy are negotiated.

1.3 Toward Plural and Value-Sensitive Family-Centered AI

Rather than designing toward a singular model of the family, we propose thinking about plural family-centered AI: systems that can accommodate different family structures, relationships, values, and material circumstances without positioning one configuration as normative. Value Sensitive Design (VSD) offers one useful foundation for this approach. VSD provides methods for systematically examining stakeholders and human values throughout technological design [8]. We therefore suggest four directions for family-centered AI research and design.

- **Design relationships rather than fixed roles:** AI systems should avoid assuming that every household can be represented through predefined “parent” and “child” roles. Researchers

should investigate who actually participates in care, technological support, and decision-making and explore mechanisms through which families can configure trusted relationships.

- **Support negotiated visibility rather than default surveillance:** Family-centered safety systems should explore graduated and context-sensitive mechanisms for information sharing. Young people might, for example, identify trusted individuals, specify categories of information they are willing to share, or determine when assistance should be requested rather than having all activity automatically exposed to caregivers.
- **Treat infrastructure as a design dimension:** Shared devices, intermittent connectivity, varying digital literacy, and uneven technological resources should be considered during the conceptualization of family-centered AI rather than treated solely as deployment limitations.
- **Elicit family values rather than presuming them:** Researchers should examine how family members themselves understand care, privacy, responsibility, authority, and autonomy and how these values are negotiated when they conflict. Family-centered AI should support these negotiations rather than encode a predetermined hierarchy among them.

Ultimately, we argue that family-centered AI should not be defined simply as AI designed for multiple members of a household. Rather, it should be understood as AI designed with sensitivity to the relationships through which families negotiate care, authority, autonomy, resources, and responsibility.

1.4 Questions We Bring to the Workshop

Our perspective brings together youth online safety, research with global south insights, and Value Sensitive Design to contribute to emerging conversations about what it means to design AI for families. In particular, our experiences studying youth online safety in Nigeria lead us to three questions that we hope to explore with the workshop community:

- (1) Whose conception of “family” becomes encoded when we design family-centered AI?
- (2) How can family-centered AI support care and protection without positioning surveillance as the default mechanism through which families keep young people safe?
- (3) How can we design AI that remains responsive to variation in family structures, values, technological resources, and sociocultural contexts rather than treating one configuration of family life as universal?

Answering these questions requires moving beyond simply placing AI within existing family structures. It requires examining how AI itself may redistribute information, authority, care, and responsibility among family members. As AI becomes increasingly embedded within everyday life, family-centered design presents an opportunity not only to ask how families might use AI, but also to critically examine what kinds of families our AI systems imagine in the first place.

2 Author Biographies

Ozioma C. Oguine is a PhD student in Computer Science and Engineering at the University of Notre Dame. As a BIPOC researcher living with disability, his research investigates how online and emerging technologies can be designed to better protect and empower marginalized and vulnerable users, particularly in relation to online safety & privacy, digital well-being, and participation in technology design and governance.

Dr. Karla Badillo-Uruqiola (Dr. KB) is a Clare Boothe Luce Assistant Professor of Computing Science and Engineering at the University of Notre Dame. Her expertise lies at the intersection of Human-Computer Interaction, Critical Design for Computing, and Adolescent Online Safety, specifically for vulnerable and marginalized youth. Her research has won Best Paper (top 1%), Best Paper Honorable Mention (top 5%), Best Poster, and Impact Recognition awards at top HCI venues.

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